

Module 8 - Routing and Advance Routing

Network Access, Basic and Advance routing concept & switching concept

Ans:

A switch is a device that connects many devices in the same LAN (local area network) and sends data from one device to another.

It is an *intelligent* device that works on the **Data Link layer** and uses **MAC addresses** to forward data properly.

Types of switches:

- Managed Switch
- Unmanaged switch

Managed Switch:

These switches offer more control and configuration options so you can customize your network as per requirement.

Unmanaged Switch:

Just plug-and-play type switches. No configuration is needed—just connect and it works automatically.

2 > Switch boot sequence

Ans:

1. The switch performs a Power-On Self Test (POST) stored in ROM.
 2. Then, the bootloader program runs, which is also stored in ROM.
 3. The bootloader does CPU initialization (registers, memory mapping, etc).
 4. It initializes the flash file system.
 5. Then, the IOS (Operating System) is loaded into memory and control is handed over to it.
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3 > Three Ways to Access the Switch CLI

Ans:

1. Console Access
2. Telnet (In-band)
3. SSH (Secure Shell - In-band)

Console Access:

You connect a computer/laptop directly to the switch using a console cable (RJ-45 or USB to Serial).

Telnet:

Access the CLI remotely through network using the **Telnet** protocol (requires setting IP on switch first).

SSH:

Works like telnet, but it's **secure and encrypted** for safer remote access.

4 > Cisco IOS Explained + Configuration

Ans:

Cisco IOS is the Operating System used in Cisco switches and routers. It:

- Gives access to CLI for configuration and monitoring
 - Implements protocols like OSPF, VLAN, EIGRP, ACLs etc
 - Controls hardware like interfaces, memory, routing/switching engine
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Sample Configuration:

```
Switch> enable
```

```
Switch# configure terminal
```

```
Switch(config)# hostname MySwltch
```

```
MySwltch(config)# enable secret Pass123
```

```
MySwltch(config)# interface vlan 1
```

```
MySwltch(config-if)# ip address 192.168.1.10 255.255.255.0
```

```
MySwitch(config-if)# no shutdown
MySwitch(config)# ip default-gateway 192.168.1.1
MySwitch(config)# line console 0
MySwitch(config-line)# password ConsolePass
MySwitch(config-line)# login
MySwitch(config)# line vty 0 4
MySwitch(config-line)# password VtyPass
MySwitch(config-line)# login
MySwitch(config-line)# transport input ssh
MySwitch(config)# ip domain-name example.com
MySwitch(config)# crypto key generate rsa
*Choose key size (e.g. 1024 or 2048 bits)*
MySwitch(config)# username admin secret AdminPass
MySwitch(config-line)# login local
MySwitch# write memory
MySwitch# copy running-config startup-config
```

5 > Switch Port

Ans:

A switch port is a **physical or logical point** where you connect devices like PC, printer, or access points using Ethernet cable.

It acts like a socket to plug-in and communicate in the network.

Why Important?

- **Connectivity:** It's the point of entry for all devices in LAN.
- **Segmentation:** VLANs allow ports to isolate traffic logically.
- **Security:** Port security can restrict which MAC addresses are allowed to connect.